

Biophysics, with its Computational Engine

June 2026

Abstract

Biophysics is what happens when you take the statistical mechanics you already know and turn it loose on warm, wet, crowded, fluctuating matter that happens to be alive. This report assumes you are comfortable with thermodynamics and the Boltzmann distribution, with stochastic processes at the level of diffusion, and with modern machine learning — but not with biology. We start from the physical regime that makes living matter distinctive (the dominance of $k_{\text{B}}T$, low Reynolds number, far-from-equilibrium operation), build up the statistical mechanics of macromolecules and their free-energy landscapes, survey the molecular machines and membranes that physics has to explain, and then devote the second half to the *computational engine*: molecular dynamics, enhanced sampling and free-energy estimation, coarse-graining, and the deep-learning revolution in structure prediction and protein design. Throughout, the connective tissue to your ML work — equivariant networks, generative models, sampling — is made explicit.

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1 The Physical Regime of Life

Biophysics is not “physics topics that biologists find interesting”; it is the physics of a specific and unusual regime of matter. Three facts define it.

1. Thermal energy is the natural unit, and it is large. At body temperature $T \approx 310$ K,

$$k_B T \approx 4.3 \text{ pN} \cdot \text{nm} \approx 0.6 \text{ kcal/mol} \approx 26 \text{ meV}.$$

Every relevant interaction — a hydrogen bond ($\sim 2\text{--}5 k_B T$), the energy that holds a folded protein together ($\sim 10\text{--}20 k_B T$ *net*, out of hundreds of competing terms), the chemical energy released by hydrolyzing one ATP molecule ($\sim 20 k_B T$) — is within a small multiple of $k_B T$. Consequence: biomolecules are perpetually kicked by thermal noise of the same magnitude as the forces holding them together. Life operates at the edge of thermal chaos, not safely above it. This is why *ensembles and probabilities*, not single trajectories, are the right description — exactly the statistical-mechanics mindset.

2. The world is viscous, not inertial (low Reynolds number). For a bacterium of size $a \sim 1 \mu\text{m}$ swimming at $v \sim 10 \mu\text{m/s}$ in water, the Reynolds number is

$$\text{Re} = \frac{\rho v a}{\eta} \sim 10^{-4}.$$

Inertia is utterly negligible; a swimming microbe coasts for less than an ångström when it stops paddling. Newton’s $F = ma$ is effectively replaced by the overdamped relation $F = \gamma v$ (force $\propto$ velocity). Purcell’s famous lesson: at low Re, reciprocal motion (a back-and-forth flap) produces *zero* net displacement, which is why bacterial flagella rotate rather than wave.

3. Life is far from equilibrium. A living cell continuously burns free energy (ATP, ion gradients) to stay ordered. Equilibrium statistical mechanics describes the *molecules* on short timescales but the cell as a whole is a driven, dissipative steady state. This is the open frontier — nonequilibrium statistical mechanics — and where the cleanest physics is still being written (fluctuation theorems, stochastic thermodynamics).

The governing equation of the small, wet world

Most of molecular biophysics is the **Langevin / overdamped dynamics** of a particle in a potential U :

$$\gamma \dot{\mathbf{x}} = -\nabla U(\mathbf{x}) + \boldsymbol{\xi}(t), \quad \langle \xi_i(t) \xi_j(t') \rangle = 2\gamma k_B T \delta_{ij} \delta(t - t').$$

The friction γ and the noise $\boldsymbol{\xi}$ are linked by the fluctuation–dissipation theorem; their ratio fixes the diffusion constant via Einstein’s relation $D = k_B T / \gamma$. The corresponding distribution of positions relaxes to Boltzmann, $p(\mathbf{x}) \propto e^{-U(\mathbf{x})/k_B T}$, by the Fokker–Planck equation. If you understand this line, you understand the skeleton of the entire field — and, not coincidentally, the continuous-time limit of diffusion generative models.

2 Macromolecules and the Forces that Fold Them

The principal actors are **proteins** (polymers of ~ 20 amino-acid types, the cell’s machines and structure) and **nucleic acids** (DNA/RNA, the information polymers). A protein is a linear chain

that, under the right conditions, folds into a specific 3D structure that determines its function — the central structure–function paradigm.

The forces are all familiar electromagnetism, dressed for the aqueous environment:

- **Hydrogen bonds** — directional, $\sim 2\text{--}5 k_B T$ each; organize the secondary structure (α -helices, β -sheets).
- **Van der Waals / steric packing** — weak and short-ranged but numerous; tight core packing.
- **Electrostatics**, heavily *screened* by water ($\epsilon \approx 80$) and mobile ions (Debye length ~ 1 nm at physiological salt). Coulomb’s law with a Debye–Hückel exponential cutoff.
- **The hydrophobic effect** — the dominant folding driver, and the subtle one: it is *entropic*. Nonpolar side chains cluster not because they attract each other but because exposing them forces water into ordered cages, costing entropy. Burying them frees that water. Folding is largely the system maximizing *water’s* entropy.

Folding is a free-energy competition, won by a sliver

The folded state is favored by enthalpy (bonds, packing) but disfavored by chain entropy (a folded chain has far fewer conformations). The net stability

$$\Delta G_{\text{fold}} = \Delta H - T\Delta S$$

is a small difference of large numbers — typically only $\sim 5\text{--}15 k_B T$. Proteins are marginally stable *by design*: stable enough to function, loose enough to move, be regulated, and be degraded.

3 Statistical Mechanics of Conformations: the Energy Landscape

The unifying picture of the field is the **free-energy landscape**: a function $F(\mathbf{x})$ over conformational coordinates, where $\mathbf{x}$ might be all $3N$ atomic coordinates or a few collective “reaction coordinates.” Equilibrium populations are Boltzmann, $p(\mathbf{x}) \propto e^{-F(\mathbf{x})/k_B T}$, and dynamics is the overdamped motion of §1 across this landscape.

Two landmark ideas:

- **The folding funnel.** Levinthal’s paradox: a 100-residue chain with even 3 states per residue has $3^{100} \approx 10^{47}$ conformations; random search would take longer than the age of the universe, yet folding takes microseconds to seconds. Resolution: the landscape is not flat but *funneled* — biased globally toward the native state — so the chain is steered downhill, never searching blindly. The native structure sits at the bottom of a rugged funnel.
- **Rates as barrier crossing (Kramers).** Transitions between metastable states (folded $\leftrightarrow$ unfolded, open $\leftrightarrow$ closed) are thermally activated barrier crossings with rate $k \sim k_0 e^{-\Delta F^\ddagger/k_B T}$. The exponential sensitivity to barrier height $\Delta F^\ddagger$ is why a few $k_B T$ separates a millisecond process from an hour-long one, and why *sampling these transitions is the central computational difficulty* (§6.2).

4 Polymers, Mechanics, and Single Molecules

Stripped of chemical detail, a long biomolecule is a **polymer**, and polymer physics gives quantitative laws. A DNA molecule resists bending with a *persistence length* ℓ_p (the length over which the chain “forgets” its direction): $\ell_p \approx 50$ nm for double-stranded DNA, but only ~ 1 nm for a floppy protein chain. The **worm-like chain** model predicts the force needed to stretch such a polymer:

$$\frac{F\ell_p}{k_B T} \approx \frac{1}{4} \left(1 - \frac{x}{L}\right)^{-2} - \frac{1}{4} + \frac{x}{L},$$

where x/L is the fractional extension. Crucially, the restoring force is *entropic*: stretching a chain straightens it, reducing its conformational entropy, and the chain pulls back to reclaim disorder — a polymer is an entropic spring.

This is not just theory: **single-molecule experiments** (optical/magnetic tweezers, AFM, single-molecule FRET) now grab individual molecules and measure force–extension curves, watch a single motor step, or clock a single folding event, directly testing these landscapes. The natural force unit, fittingly, is the piconewton: $k_B T/\ell \sim 4$ pN at nanometer scales.

5 Molecular Machines, Membranes, and Channels

Molecular motors (myosin on actin, kinesin on microtubules, the rotary motors ATP synthase and the flagellar motor) convert chemical free energy into directed motion. The physics puzzle: how do you get *directed* work out of a machine bathed in symmetric thermal noise? The answer is the **Brownian ratchet** — the motor does not push against thermal kicks but rectifies them, using ATP binding/hydrolysis to break detailed balance and bias diffusion in one direction. They are nonequilibrium devices operating near the thermodynamic efficiency limit.

Membranes are self-assembled lipid bilayers — two-dimensional fluids whose existence is again driven by the hydrophobic effect (tails hide from water, heads face it). Their elasticity (bending modulus, tension) is described by the Helfrich Hamiltonian, a piece of differential-geometric physics.

Ion channels and the membrane potential are where biophysics meets electrophysiology. A concentration gradient of an ion across a membrane sets up the **Nernst potential**

$$V_{\text{eq}} = \frac{k_B T}{ze} \ln \frac{[\text{ion}]_{\text{out}}}{[\text{ion}]_{\text{in}}},$$

and the celebrated **Hodgkin–Huxley** model describes the nerve action potential as a nonlinear dynamical system of voltage-gated conductances — arguably the first great quantitative, predictive model in biology, and pure applied physics (RC circuits plus stochastic gating).

6 The Computational Engine

Analytic models take you remarkably far, but a folded protein with explicit water is a 10^4 – 10^6 -atom interacting system with no closed-form solution. Computation is not an add-on to biophysics — since the 1970s it has been a third pillar alongside theory and experiment. The rest of this report is the computational toolkit.

6.1 Molecular dynamics (MD)

The workhorse. Treat every atom as a classical point mass (quantum effects on nuclei are usually negligible at $k_B T$), assign a **force field** — a parameterized potential energy

$$\begin{aligned}
 U = & \underbrace{\sum_{\text{bonds}} \frac{1}{2} k_b (r - r_0)^2 + \sum_{\text{angles}} \frac{1}{2} k_\theta (\theta - \theta_0)^2 + \sum_{\text{dihedrals}} V_n [1 + \cos(n\phi - \delta)]}_{\text{bonded}} \\
 & + \underbrace{\sum_{i < j} 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] + \sum_{i < j} \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}}}_{\text{nonbonded: Lennard-Jones + Coulomb}}
 \end{aligned}$$

— and integrate Newton’s equations (or Langevin dynamics with a thermostat) with the velocity-Verlet algorithm. Standard force fields are AMBER, CHARMM, OPLS; common engines are GROMACS, OpenMM, NAMD, and Amber.

The timescale problem — the field’s defining frustration

Chemical bonds vibrate at ~ 10 fs, so numerical stability forces a timestep of $\Delta t \approx 1$ fs to 2 fs. But protein folding, large conformational changes, and ligand binding happen on μ s–s timescales. Reaching 1 ms thus needs $\sim 10^{12}$ steps. This 10^9 – $10^{12} \times$ gap between timestep and phenomenon is *the* central computational obstacle. Three families of responses follow.

6.2 Enhanced sampling and free-energy estimation

Brute-force MD wastes essentially all its time rattling inside one free-energy basin, never crossing the Kramers barriers that matter. Enhanced-sampling methods buy access to the rare transitions and to free-energy *differences* (the quantities that predict binding affinities and stabilities):

- **Umbrella sampling:** add biasing harmonic potentials along a chosen reaction coordinate to flatten barriers, then unbias with WHAM/MBAR to recover the true $F(\xi)$.
- **Metadynamics:** adaptively deposit Gaussian “hills” to fill basins already visited, pushing the system to explore; the accumulated bias *is* the negative free energy.
- **Replica exchange (parallel tempering):** run many copies at different temperatures and swap them, letting hot replicas hop barriers and cool ones refine — a Monte Carlo move in temperature space.
- **Free-energy perturbation / thermodynamic integration:** compute ΔG between two states (e.g. drug A vs. drug B binding) via the Zwanzig relation $\Delta G = -k_B T \ln \langle e^{-\Delta U/k_B T} \rangle$. This is the engine of computational drug discovery.
- **Jarzynski’s equality,** $\langle e^{-W/k_B T} \rangle = e^{-\Delta G/k_B T}$, a nonequilibrium fluctuation theorem that extracts *equilibrium* free energies from *irreversible* pulling — one of the most beautiful results in the field.

6.3 Coarse-graining and Markov state models

Coarse-graining throws away atomistic detail — lumping groups of atoms into single beads (e.g. the MARTINI model) — to reach larger systems and longer times, at the cost of carefully refitting the effective potential to reproduce the quantity you care about. The inverse problem (which fine-grained details to keep) is deep and increasingly attacked with ML.

Markov state models (MSMs) take the opposite tack: run many *short* MD trajectories, cluster the configurations into discrete states, and estimate the transition-probability matrix between them. The leading eigenvectors of that matrix reveal the slow collective processes (folding, the dominant conformational switches) without ever simulating a single long trajectory — stitching statistics together from parallel short runs. This is spectral analysis of a stochastic operator, squarely in your linear-algebra comfort zone.

7 The Deep-Learning Revolution

Since roughly 2020, machine learning has changed which biophysics problems are tractable. This is the part nearest your own work.

Structure prediction: AlphaFold and after

The 50-year “protein folding problem” — predict 3D structure from sequence — was substantially solved as a *prediction* task by **AlphaFold2** (2021), which reached near-experimental accuracy at CASP14. It does *not* simulate folding physics; it learns the sequence→structure map from the Protein Data Bank, exploiting coevolutionary signal in multiple-sequence alignments via attention (the Evoformer) and an SE(3)-aware structure module. **AlphaFold3** (2024) and peers (RoseTTAFold, ESMFold) extended this to complexes, nucleic acids, and ligands with diffusion-based structure generation. Caveat worth internalizing: these predict a *single dominant structure*, not the Boltzmann *ensemble* or the folding pathway — so they complement, not replace, the physics of §3–§5.

Generative design and neural force fields

- **Protein design** runs the map backwards: *generate* a sequence/structure with a desired function. **RFdiffusion** and related models are denoising diffusion models over backbone coordinates — the same diffusion framework you know, operating on SE(3) rigid-body frames — paired with inverse-folding networks (ProteinMPNN) that choose a sequence to realize the designed shape. This is the computational backbone of modern *de novo* protein and binder design.
- **Machine-learned interatomic potentials** (NequIP, MACE, Allegro, and large pre-trained models) fit the energy surface from quantum-chemistry data, delivering near-*ab initio* accuracy at force-field cost — shrinking the MD accuracy/speed tradeoff.
- **Boltzmann generators** and flow/diffusion samplers aim to draw *equilibrium* samples $p \propto e^{-U/k_B T}$ directly, attacking the sampling problem (§6.2) with a learned change of variables instead of long dynamics — conceptually, learning to teleport across barriers.
- **Cryo-EM reconstruction** now uses deep generative models (e.g. cryoDRGN) to recover not one structure but the continuous *distribution* of conformations from noisy single-particle images — an inverse problem married to a generative prior.

The thread back to your algebra report

Nearly every successful molecular network is SE(3)- or SO(3)-**equivariant**: rotate the input molecule and the predicted forces, fields, and coordinates rotate with it. This is the representation theory of the rotation group — spherical harmonics as irreps, tensor products via Clebsch–Gordan coefficients — turned into architecture. The symmetry-as-inductive-bias principle is not optional here; it is what makes these models data-efficient enough to learn physics. Biophysics is where the abstract algebra, the statistical mechanics, and the ML in your prior reports all converge.

8 The Map of the Field

Scale / object	Governing physics	Computational tool
Atoms, bonds	quantum chemistry, $k_B T$ thermal	QM/DFT, ML potentials
Single molecule	statistical mechanics, polymer physics	MD, enhanced sampling, MSMs
Folding / binding	free-energy landscapes, Kramers	FEP/TI, metadynamics, AlphaFold
Machines / motors	nonequilibrium thermodynamics	MD + driving, kinetic models
Membranes / channels	elasticity, electrostatics, RC dynamics	coarse-grained MD, Hodgkin–Huxley
Design (inverse problem)	sequence↔structure map	diffusion models, inverse folding

9 Things Worth Getting Right Early

- *Entropy is half the story, often the dominant half.* The hydrophobic effect and polymer elasticity are *entropic* forces. Coming from energy-minimization intuition (and from ML loss landscapes), it is easy to forget that $-T\Delta S$ frequently wins.
- *A predicted structure is not an ensemble.* Boltzmann weights, multiple metastable states, and pathways carry the biology; a single AlphaFold structure is the mode, not the distribution. Don’t mistake one for the other.
- *Sampling, not energy evaluation, is usually the bottleneck.* You can often score a configuration cheaply; the hard part is visiting the right configurations with the right frequencies. This reframes most “why is this slow” questions.
- *Force fields are models with error bars.* MD results are only as good as the parameterized potential; a beautiful trajectory can be quantitatively wrong. Validate against experiment.

10 Where to Go Next

- **Phillips, Kondev, Theriot & Garcia, *Physical Biology of the Cell*** — the modern standard; physics-first, quantitative, and the best single match for your background.

- **Nelson**, *Biological Physics* — excellent on the statistical-mechanics and single-molecule foundations (§1–§5 here).
- **Frenkel & Smit**, *Understanding Molecular Simulation* — the canonical text for MD/MC, free-energy methods, and the algorithms of §7–§8.
- **Tuckerman**, *Statistical Mechanics: Theory and Molecular Simulation* — rigorous bridge from stat-mech to simulation.
- For the ML frontier: the **AlphaFold2/3** papers (Jumper et al. 2021; Abramson et al. 2024), **RFdiffusion** (Watson et al. 2023), **MACE** (Batatia et al. 2022), and **Boltzmann generators** (Noé et al. 2019) — each is a clean entry point given your background, and each connects directly to the generative and equivariant modeling you already do.

Biophysics is statistical mechanics with the lights on: the same Boltzmann distribution, fluctuation–dissipation, and barrier-crossing you know, applied to matter that is soft, wet, warm, driven, and astonishingly well-engineered by evolution. Its computational engine — from Verlet integration to diffusion-model protein design — is where physical law, sampling, and modern machine learning meet most productively.